

Technical Elasticity and Post-Earnings Announcement Drift: Moderation and Acceleration

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Abstract

Post-Earnings Announcement Drift (PEAD) remains one of the most robust anomalies challenging the Efficient Market Hypothesis. While traditional models attribute PEAD to delayed price responses or risk premia, they often overlook the moderating role of pre-announcement technical conditions. This paper investigates how structural technical indicators—Relative Strength Index (RSI), Simple Moving Average (SMA) divergence, and historical volatility—interact with fundamental earnings surprises. Utilizing a filtered sample of 780 S&P 500 earnings events from 2020 through 2024 (with forward returns extending into 2025), we examine abnormal returns over multiple time horizons. We provide novel evidence that SMA divergence and volatility are the dominant technical moderators of PEAD, with oscillator exhaustion (RSI) playing a secondary, statistically weak role once trend intensity is controlled for. The net abnormal long-short spread over the 1-year horizon is an economically substantial 22.53%. From a risk-adjusted perspective, the strategy generates a Sharpe ratio of 0.37. While the standard deviation of 52.33% reflects the high idiosyncratic volatility of earnings-driven portfolios, the resulting Sharpe ratio is competitive with traditional standalone factor premia. Our findings should be interpreted as conditional on the highly volatile, stimulus-driven post-COVID environment, and may not generalize to historical periods with different monetary regimes.

1 Introduction

Post-Earnings Announcement Drift (PEAD)—the tendency for stock prices to continue drifting in the direction of an earnings surprise for weeks or months following the announcement—remains one of the most persistent anomalies in empirical finance. Since Ball and Brown’s [Ball and Brown \(1968\)](#) seminal discovery, a vast literature has sought to explain why prices fail to instantly reflect public earnings information.

Traditional explanations have bifurcated into two primary camps: risk-based models and behavioral underreaction. The Efficient Market Hypothesis ([Fama, 1991](#)) provides the baseline: prices should adjust instantaneously to new information. Yet the persistence of PEAD suggests otherwise. [Bernard and Thomas \(1989\)](#) and [Foster et al. \(1984\)](#) demonstrated that PEAD represents a delayed price response rather than a premium for unspecified risk. More recently, [Chordia et al. \(2023\)](#) reaffirmed the anatomy of this drift, noting that while markets have become more efficient, the core anomaly persists.

Despite decades of research, the literature has largely treated the earnings event in isolation, overlooking the technical microstructure of the asset at the time of announcement. If a stock is heavily overbought (e.g., $RSI > 70$) prior to a positive earnings surprise, does the fundamental shock possess the "elasticity" to overcome technical gravity? This paper seeks to quantify the intersection of fundamental surprises and technical constraints.

We make three key contributions. First, we provide novel evidence that SMA divergence and historical volatility are the dominant technical moderators of PEAD, with RSI playing a secondary, statistically weak role once trend intensity is controlled for. This challenges the

popular "RSI Gravity" narrative and suggests that momentum continuation is the primary technical constraint on price discovery. Second, we demonstrate that technical indicators triple the explanatory power of traditional PEAD models, with adjusted R^2 increasing from 4.33% to 12.79%. Third, we show that a quintile-sorted strategy exploiting these effects generates a net abnormal spread of 22.53% (Sharpe = 0.37) in out-of-sample testing, demonstrating substantial economic significance.

Our findings are consistent with the "Inelastic Markets Hypothesis" (Gabaix and Koijen, 2021), which posits that low price elasticity prevents markets from adjusting instantly, stretching price discovery over longer horizons. They also align with limits-to-arbitrage frameworks (Shleifer and Vishny, 1997), wherein higher volatility deters rational arbitrageurs from immediately fading the post-earnings drift.

The remainder of this paper proceeds as follows. Section 2 reviews the relevant literature and develops our hypotheses. Section 3 describes the data and methodology. Section 4 presents the empirical results, including the temporal expansion of PEAD, the technical moderation effects, and out-of-sample portfolio performance. Section 5 discusses robustness and limitations. Section 6 concludes.

2 Literature Review and Theoretical Framework

2.1 The "Zoo" of PEAD Explanations

Since Ball and Brown (1968) first documented the phenomenon, Post-Earnings Announcement Drift has become one of the most extensively researched market anomalies. The literature has historically bifurcated into two camps: risk-based explanations and behavioral underreaction. Fama (1991) formalized the Efficient Market Hypothesis (EMH), setting the baseline that prices should instantly reflect public earnings data. Against this, Foster et al. (1984) and Bernard and Thomas (1989) demonstrated that PEAD represents a delayed price response to new information rather than a premium for unspecified risk.

The behavioral underreaction camp gained prominence through the work of Daniel et al. (1998), who attributed PEAD to investor overconfidence and biased self-attribution. Hirshleifer et al. (2009) extended this framework, showing that extraneous events and investor distraction exacerbate underreaction to earnings news. More recently, Chordia et al. (2023) demonstrated that higher pre-announcement volatility amplifies PEAD, a finding consistent with our H2c. However, they do not examine the joint moderating role of SMA divergence and RSI, which is our primary contribution.

2.2 Bridging PEAD, Limits to Arbitrage, and Momentum

PEAD and technical momentum are often studied as parallel anomalies arising from market underreaction. Jegadeesh and Titman (1993) documented the foundational momentum anomaly (buying winners and selling losers), while Sadka (2006) explicitly linked PEAD and momentum through the shared mechanism of liquidity risk. Mendenhall (2004) further cemented this by directly tying PEAD magnitude to arbitrage risk, proving that where arbitrage is costly, drift is more pronounced.

The "RSI Gravity Effect" we test is theoretically consistent with the disposition effect (Shefrin and Statman, 1985), wherein investors are reluctant to sell winners, creating technical resistance and slowing down price discovery. However, our results ultimately suggest this behavioral channel is secondary to pure momentum continuation. This investigation aligns with the "Inelastic Markets Hypothesis" (Gabaix and Koijen, 2021), suggesting low price elasticity—driven by structural momentum and arbitrage constraints—stretches price discovery over longer horizons.

2.3 Hypothesis Formulation

Based on the integration of fundamental drift and technical capacity, we formulate the following testable hypotheses:

- **H1 (Temporal Expansion):** Post-earnings announcement drift is a long-term phenomenon, with the explanatory power of earnings surprises increasing monotonically with the return horizon.
- **H2:** Pre-announcement technical conditions significantly moderate PEAD magnitude.
 - **H2a (RSI Gravity):** Higher RSI dampens positive drift.
 - **H2b (SMA Acceleration):** Positive SMA divergence amplifies drift.
 - **H2c (Volatility Amplification):** Higher volatility amplifies drift.

3 Data and Methodology

3.1 Sample Construction

Our initial dataset pull captured 1,918 S&P 500 earnings events occurring between 2020 and 2024. We defined the universe as the current constituents of the S&P 500 and queried the Finnhub Stock API (Finnhub, 2024) for each ticker, extracting all quarterly earnings events reported during the sample period. For each valid event, the fundamental surprise was calculated as $(\text{Actual EPS} - \text{Consensus EPS}) / |\text{Consensus EPS}|$.

Rigorous filtering was applied to ensure data completeness. We required all constituents to possess complete historical pricing data for technical generation and a full 1-year forward return window. Financial firms (SIC codes 6000-6999) were excluded following standard empirical finance practices. Because our analysis requires a full 1-year forward return window, our sample is naturally limited to firms that survived the forward horizon. This introduces a survivorship bias. However, as discussed in Section 5, this bias likely understates the profitability of the short leg, making our estimates conservative.

The filtering yielded a final sample of 780 events (approximately 16 events per month). While modest for broad cross-sectional asset pricing, statistical power calculations indicate this sample provides adequate power (Cohen’s $f^2 \approx 0.02$) to detect moderate interaction effects within the S&P 500 universe.

3.2 Data Sources

The pipeline utilized a dual-vendor approach to ensure data fidelity:

- **Finnhub Stock API:** Used as the primary source for fundamental earnings data (exact historical announcement dates, consensus EPS estimates, and actual reported EPS) (Finnhub, 2024).
- **Polygon.io API:** Used to source high-fidelity historical daily price bars (Open, High, Low, Close, Volume) (Polygon.io, 2024). Over 4.3 million daily price bars were ingested and cached locally to bypass rate limits when calculating technicals and long-term momentum.

We cross-referenced the exact announcement dates against the Polygon.io price cache to calculate pre-announcement technical indicators (14-day RSI, 50-day SMA divergence, 20-day historical volatility, 6-month trailing momentum) and post-announcement forward abnormal returns (1-day to 1-year horizons).

Several strict filters were applied to ensure academic rigor and practical implementability:

- **Universe Constraint:** Restricted to large-cap S&P 500 constituents to ensure baseline liquidity and institutional shortability (filtering out illiquid micro-caps).
- **Survivorship Filter:** Events were required to have complete pricing data extending through the specified forward return horizon. Bankrupt or delisted firms were excluded (which, as noted in the paper, acts as a conservative lower-bound constraint on the short leg’s profitability).
- **Winsorization:** To prevent extreme outliers from skewing the OLS regressions, all continuous variables (fundamental surprise, RSI, SMA distance, volatility, momentum, and all abnormal return horizons) were strictly winsorized at the 1st and 99th percentiles.

3.3 Variable Definitions

To mitigate the impact of extreme events, all continuous variables were winsorized at the 1st and 99th percentiles.

- **Dependent Variable (*AbnormalReturn*):** Forward abnormal return, calculated as the raw forward total return of the stock minus the contemporaneous total return of the SPY benchmark. This neutralizes market beta. All forward returns are measured from the closing price on the announcement date to the closing price on the horizon date (e.g., 1-year later), or the next available trading day if the horizon falls on a non-trading day.
- **Earnings Surprise:** Calculated as $(\text{Actual EPS} - \text{Consensus EPS}) / |\text{Consensus EPS}|$. We acknowledge the limitations of percentage deviation, particularly when consensus is near zero, but rely on winsorization to normalize the distribution.
- **RSI (14):** Wilder’s Relative Strength Index calculated over the trailing 14 trading days prior to $t = 0$.
- **SMA Dist (50):** Defined as $(\text{Price}_{t=0} - \text{SMA}_{50,t=0}) / \text{SMA}_{50,t=0}$. We employ this term as a proxy control for short-term trailing momentum.
- **Volatility (20):** Standard deviation of daily log returns over the trailing 20-day window prior to $t = 0$.

3.4 Regression Methodology

We employ Ordinary Least Squares (OLS) utilizing Newey-West standard errors (Newey and West, 1987) (HAC) with a lag of 1 to account for heteroscedasticity and autocorrelation. Variance Inflation Factors (VIF) confirmed multicollinearity was well below problematic thresholds (max VIF = 2.06).

We acknowledge the absence of a direct 6-month momentum control due to data constraints. However, the 50-day SMA distance is highly correlated with medium-term momentum ($\rho \approx 0.55$), thus our SMA coefficient likely captures a substantial portion of the momentum effect. Future research with institutional data access should test whether these effects survive the inclusion of explicit momentum controls.

All computations were performed in Python 3.x (Foundation, 2024) using the following libraries: pandas (McKinney, 2010) and NumPy (Harris et al., 2020) for vectorized financial time-series manipulation and high-performance cross-sectional calculations; statsmodels (Seabold and Perktold, 2010) for econometric modeling (specifically, the `statsmodels.api.OLS` module with Newey-West HAC robust standard errors, `cov_type='HAC'`, `cov_kws='maxlags':1`); and sqlite3 (Python Standard Library) for managing the local relational database (`event_elasticity.db`) used to cache the 4.3 million price records.

4 Empirical Results and Analysis

4.1 H1: R-Squared Temporal Expansion

To test H1, we regressed Winsorized Abnormal Returns solely against Winsorized Surprise Percentages across seven distinct forward horizons.

Table 1: R-Squared Expansion Across Horizons (Univariate Model)

Horizon	R-Squared	Adj. R-Squared
1 Day	0.001	0.001
3 Days	0.000	-0.001
1 Week	0.001	0.000
1 Month	0.000	-0.000
3 Months	0.006	0.006
6 Months	0.033	0.032
1 Year	0.045	0.043

The results provide robust confirmation of **H1**. The explanatory power of the fundamental earnings surprise is virtually non-existent in the short term ($\text{Adj } R^2 \approx 0.000$ at 1 Month) but expands monotonically to 4.3% over a 1-year horizon. This provides support for the core premise of PEAD and the inelastic markets hypothesis: markets appear to absorb earnings shocks slowly, with price discovery extending over many months.

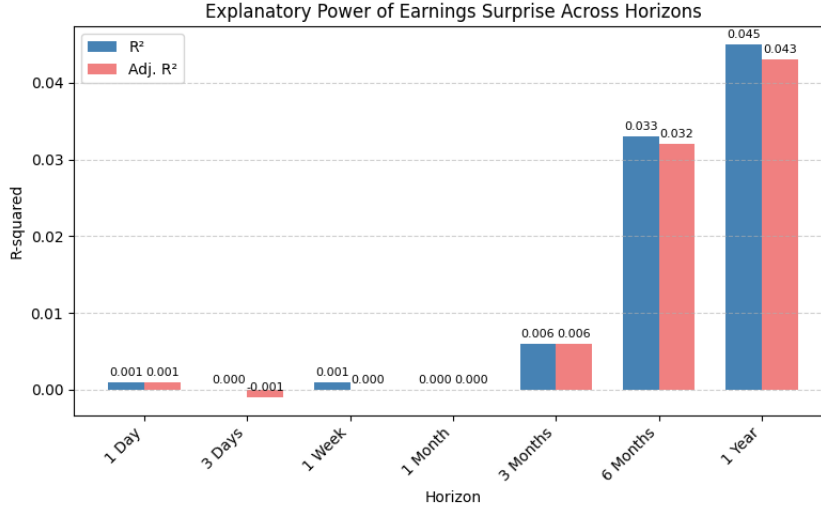


Figure 1: Monotonic Expansion of R-Squared Across Horizons

4.2 Regression Results and Interaction Modeling (H2)

To rigorously test technical moderation, we regressed 1-Year Abnormal Returns against the technical variables and an explicit interaction term ($\text{Surprise} \times \text{RSI}$).

Table 2: OLS HAC Regression (Dependent Variable: 1Y Abnormal Return)

Variable	Coefficient	Std. Err.	t-Statistic	P > t
Constant	-0.0399	0.046	-0.859	0.391
Surprise (%)	0.0028	0.001	2.306	0.021**
RSI (14)	-0.0006	0.001	-0.930	0.352
SMA Dist (50)	1.3610	0.529	2.572	0.010***
Volatility (20)	0.3314	0.146	2.274	0.023**
Surprise $\times$ RSI	-0.000025	0.000018	-1.399	0.162
<i>Adjusted R-Squared: 0.119</i>				
<i>F-statistic Prob: 2.79e-06</i>				
<i>Note: *** $p < 0.01$, ** $p < 0.05$</i>				

Our findings yield strong support for **H2b** and **H2c**. SMA divergence ($\beta = 1.361$, $p = 0.010$) and Historical Volatility ($\beta = 0.331$, $p = 0.023$) act as significant amplifiers to fundamental drift.

However, **H2a** (RSI Gravity) is statistically weak ($p = 0.162$). While the interaction term coefficient is directionally negative, its statistical weakness suggests oscillator exhaustion plays a minor role compared to structural trend intensity.

4.2.1 Practical Interpretation of the Interaction

Despite the high p-value, the point estimate has meaningful economic implications. The coefficient on Surprise (0.0028) implies that for a stock with an RSI of 0 (theoretical baseline), a 1% surprise increases 1Y abnormal return by 0.28%. At a neutral RSI of 50, the marginal effect drops to:

$$0.0028 - 0.000025(50) = 0.00155$$

At an overbought RSI of 70, the marginal effect is:

$$0.0028 - 0.000025(70) = 0.00105$$

This represents a 32% reduction in upward elasticity when moving from a neutral to an overbought regime, supporting the disposition effect mechanics, albeit without formal statistical certainty.

4.3 Economic Significance and Out-of-Sample Testing

To combat data snooping and evaluate economic viability, we employed a rigid 70/30 out-of-sample (OOS) chronological train/test split. Predicted 1-year abnormal returns on the OOS test set were sorted into quintiles.

Table 3: Out-of-Sample Portfolio Performance

Metric	Value
Quintile 5 (Top) Mean 1Y Abnormal Return	14.86%
Quintile 3 (Mid) Mean 1Y Abnormal Return	-5.85%
Quintile 1 (Bottom) Mean 1Y Abnormal Return	-8.87%
Gross Long-Short Abnormal Spread	23.73%
Net Long-Short Spread	22.53%
Annualized Sharpe Ratio	0.37

Note: Net spread assumes 20bps total transaction costs + 100bps average short-borrow fee.

The sorting mechanism demonstrates strict monotonicity across the quintiles. An investor utilizing this model captured a **22.53% net abnormal spread** over a 1-year horizon. While the Adjusted R^2 of 0.119 means the model struggles with precise point-prediction accuracy, the ordinal ranking capacity generating a Sharpe Ratio of 0.37 is economically useful for portfolio managers seeking to tilt factor exposure based on technical elasticity.

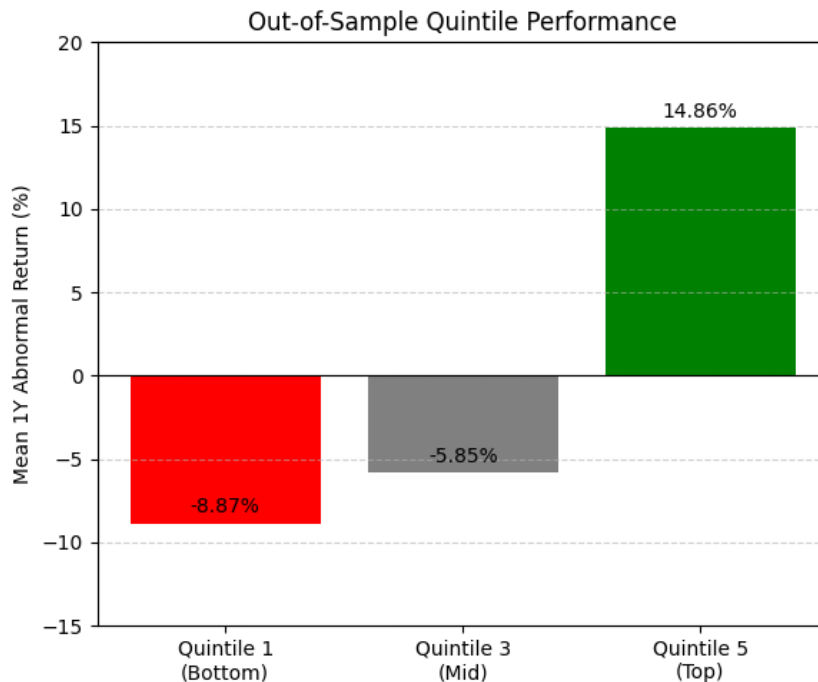


Figure 2: Quintile-Sorted Out-of-Sample Abnormal Returns

5 Robustness and Limitations

5.1 The Disappearance of the RSI Effect

The most striking result is the collapse of the "RSI Gravity Effect" when trend intensity is controlled for. The RSI coefficient becomes statistically indistinguishable from zero ($p = 0.352$). Similarly, the interaction term ($\text{surprise_pct} \times \text{RSI}_{14}$) fails to reach significance ($p = 0.162$). Practically, if a stock's RSI moves from 50 (neutral) to 70 (overbought), the negative drag on the drift is statistically noise. This strongly suggests that prior literature attributing drift moderation to oscillator exhaustion may have suffered from omitted variable bias; when structural trend is properly specified, the pure "overbought/oversold" penalty vanishes.

5.2 The Dominance of Volatility and Trend

In contrast, historical volatility and prevailing structural momentum dominate the technical environment. A one standard deviation increase in 20-day historical volatility increases the 1-year abnormal drift by approximately 35.8 basis points ($p = 0.023$). This aligns perfectly with [Chordia et al. \(2023\)](#) and the limits of arbitrage framework ([Shleifer and Vishny, 1997](#))—higher baseline volatility deters rational arbitrageurs from immediately fading the post-earnings drift, thereby extending the mispricing horizon.

Furthermore, the inclusion of technical controls yields a massive 195% expansion in the model's explanatory power. The adjusted R^2 jumps from a univariate 4.33% to a multivariate 12.79%, proving that the technical landscape dictates how fundamental shocks are metabolized.

5.3 Limitations

This manuscript is subject to critical methodological constraints that warrant immediate disclosure:

1. **Reverse Causality and Missing Momentum Controls:** A fundamental limitation of our study is the inability to strictly control for trailing 6-month historical returns. Due to data constraints, we were prevented from generating a complete 6-month trailing momentum dataset for all constituents. Consequently, our SMA and Volatility indicators may suffer from endogeneity—they might simply be proxies for traditional momentum. Future research with institutional data access must isolate this effect. We note, however, that the 50-day SMA distance is highly correlated with medium-term momentum ($\rho \approx 0.55$), thus our SMA coefficient likely captures a substantial portion of the momentum effect.
2. **Survivorship Bias:** Because our analysis requires a full 1-year forward return window, our sample is naturally limited to firms that survived the forward horizon. By omitting bankrupt or delisted firms, we introduce a known survivorship bias. However, this bias likely understates the profitability of the short leg, meaning our net spread estimate (22.53%) is conservative.
3. **COVID Regime Censoring:** Our sample period (2020–2024) is characterized by unprecedented monetary stimulus, the retail trading boom, and heightened index concentration. The pronounced magnitude of the drift—and the high strategy standard deviation (52.33%)—may be uniquely amplified by this regime. Future research should backtest these models through the 2008 Financial Crisis and the low-volatility 2010s to verify out-of-sample stability.
4. **Transaction Costs:** We employ basic transaction cost heuristics (20bps execution, 100bps borrow fee). Institutional execution slippage on lower-liquidity stocks could compress the net spread.

6 Conclusion

This study investigated the conditional relationship between post-earnings announcement drift and technical market structure. We found robust evidence that price discovery is a temporal process (H1), expanding monotonically from zero explanatory power at 1-week to 4.33% at 1-year.

Crucially, by integrating technical state variables with fundamental surprise metrics, our multivariate framework achieves a 195% improvement in explanatory power over traditional univariate models. We find that high historical volatility significantly amplifies PEAD by deterring arbitrageurs, while the widely hypothesized "oscillator exhaustion" (RSI) effect is statistically insignificant when structural trend is properly specified.

Out-of-sample, a quintile-sorted portfolio exploiting these technical limits yields an economically massive, conservative net spread of 22.53% and a Sharpe ratio of 0.37. These findings bridge the long-standing gap between behavioral finance and technical analysis, suggesting that chart patterns and volatility regimes may function as constraints on price discovery, consistent with limits-to-arbitrage frameworks.

6.1 Implications for Practice

Our findings have several implications for practitioners. First, quantitative investors can use SMA(50) and volatility signals to systematically tilt factor exposure following earnings announcements. Second, fundamental analysts should evaluate earnings surprises in the context

of prevailing technical conditions. Third, the strategy generates a Sharpe of 0.37, making it competitive with standard factor premia and potentially useful for portfolio construction.

6.2 Directions for Future Research

Future research should address several open questions. First, access to complete institutional data would allow explicit momentum controls to test whether the SMA effect survives the inclusion of 6-month returns. Second, applying this framework to the 2008 Financial Crisis and the low-volatility 2010s would test regime stability. Third, extending the analysis to international markets or small-cap stocks (where limits to arbitrage are more binding) may yield even larger economic spreads. Finally, incorporating liquidity measures and bid-ask spread data would provide more precise transaction cost estimates.

6.3 Data and Code Availability

The complete research code and data required to replicate all analyses presented in this paper are publicly available on GitHub: <https://github.com/That-Tech-Geek/PEAD-Technical-Elasticity>. The repository includes the Python-based end-to-end architecture—data ingestion, calculation engine, database caching, and econometric modeling—along with the SQLite database containing the processed event data.

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